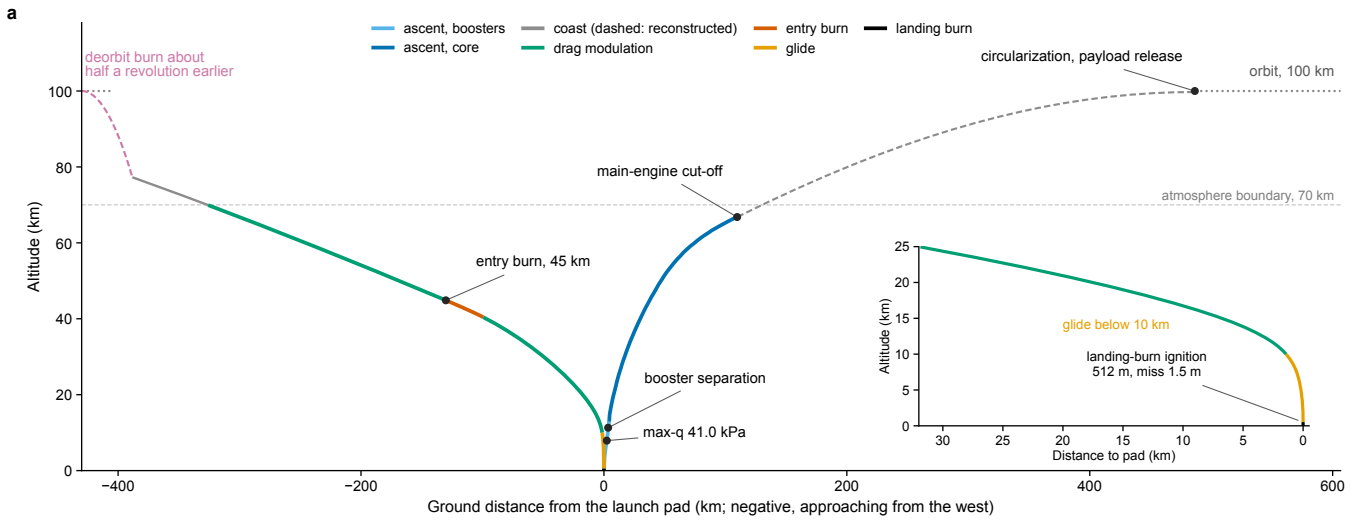
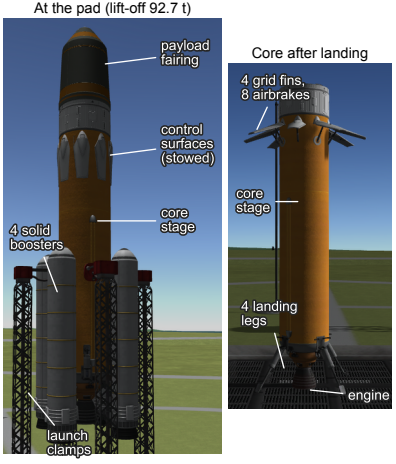
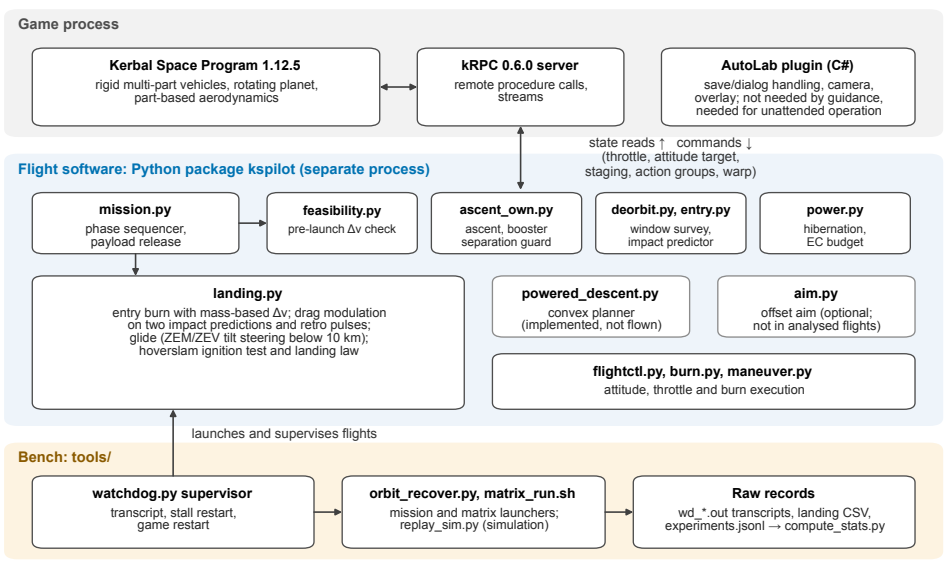

a
ascent, boosters
ascent, core
coast (dashed: reconstructed)
drag modulation
entry burn
glide
landing burn
deorbit burn about
half a revolution earlier
circularization, payload release
orbit, 100 km
main-engine cut-off
atmosphere boundary, 70 km
entry burn, 45 km
booster separation
max-q 41.0 kPa
glide below 10 km
landing-burn ignition
512 m, miss 1.5 m
Altitude (km)
Distance to pad (km)
Ground distance from the launch pad (km; negative, approaching from the west)


b
At the pad (lift-off 92.7 t)
payload
fairing
control
surfaces
(stowed)
core
stage
4 solid
boosters
launch
clamps
Core after landing
4 grid fins,
8 airbrakes
core
stage
4 landing
legs
engine


c
Game process
Kerbal Space Program 1.12.5
rigid multi-part vehicles, rotating planet, part-based aerodynamics
kRPC 0.6.0 server
remote procedure calls, streams
AutoLab plugin (C#)
save/dialog handling, camera, overlay; not needed by guidance, needed for unattended operation
state reads ↑ commands ↓
(throttle, attitude target, staging, action groups, warp)
Flight software: Python package kspilot (separate process)
mission.py
phase sequencer, payload release
feasibility.py
pre-launch Δv check
ascent_own.py
ascent, booster separation guard
deorbit.py, entry.py
window survey, impact predictor
power.py
hibernation, EC budget
landing.py
entry burn with mass-based Δv; drag modulation on two impact predictions and retro pulses; glide (ZEM/ZEV tilt steering below 10 km); hoverslam ignition test and landing law
powered_descent.py
convex planner (implemented, not flown)
aim.py
offset aim (optional; not in analysed flights)
flightctl.py, burn.py, maneuver.py
attitude, throttle and burn execution
launches and supervises flights
Bench: tools/
watchdog.py supervisor
transcript, stall restart, game restart
orbit_recover.py, matrix_run.sh
mission and matrix launchers; replay_sim.py (simulation)
Raw records
wd_*.out transcripts, landing CSV, experiments.jsonl → compute_stats.py